

Microscale Phase Change Causes Spectral Break in Turbulent Clouds: Evidence from DNS

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Key Points:

- This manuscript is under preparation and does not contain introduction, model setup, and conclusion yet. **Only key findings are pointed out.**
- The model setup and governing equations can be found in **Sharfuddin et al. (2026)**.

Abstract

[To be done later]

1 Introduction

[To be done later]

2 Governing Equations

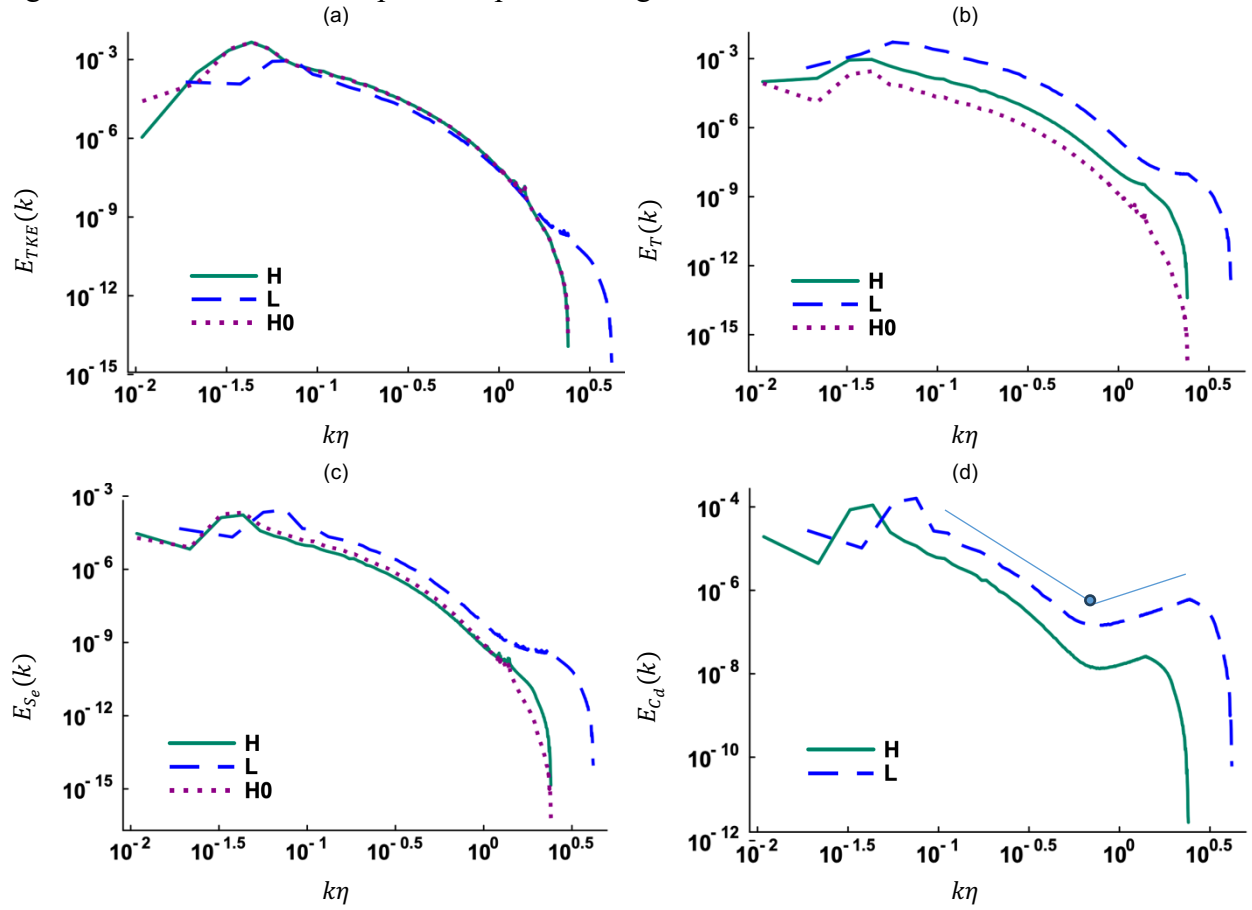
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3 Numerical Details

[To be done later]

4 Key Results and Discussion

We conduct particle-based direct numerical simulation analysis of scale separation in turbulent atmospheric clouds. The results shown here used 256^3 grid points and a 0.512^3 domain volume. Case H refers to high turbulence, Case L refers to low turbulence, and Case H0 refers to high turbulence but excludes particles/phase change.



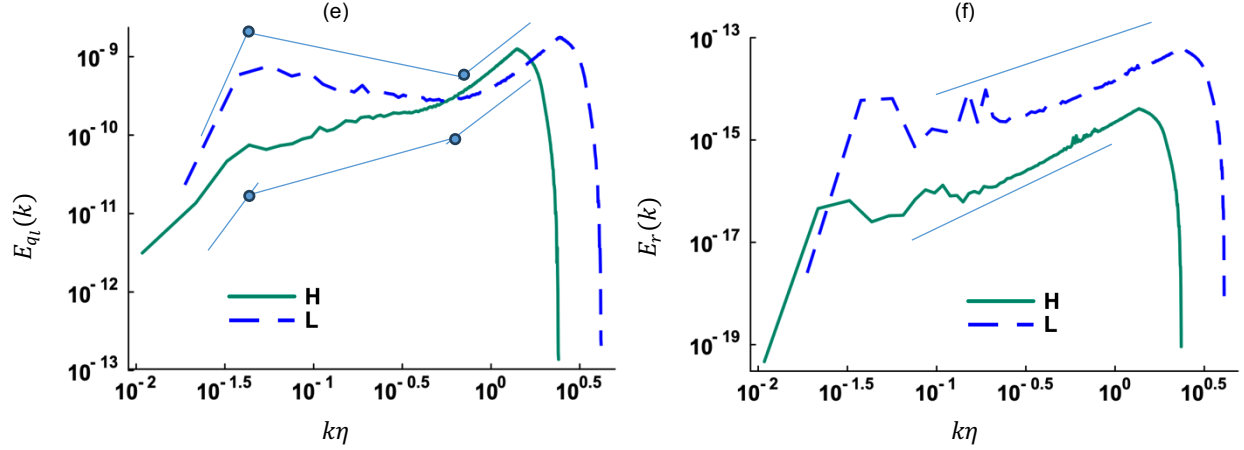


Figure 1. (a) Energy spectra of (a) TKE, (b) temperature (T), (c) environmental supersaturation (S_e), (d) mass exchange term (C_d), (e) liquid water mixing ratio (q_l), and (f) particle radius. Note that particle radius is a Lagrangian quantity while the other quantities are Eulerian. The slopes of the straight lines intersect and create spectral break.

Summary of findings in Figure 1

- 1) The TKE spectra (Fig. 1(a)) are identical for Cases H and H0, suggesting that cloud droplets do not influence the flow velocity. The E_{TKE} is higher at large scales for Cases H and H0 as expected in low-wavenumber forcing. The E_{TKE} profiles converge at intermediate scales. But there is lack of convergence at highest wavenumbers because we use 256^3 grid points for both turbulence levels, making the $k_{max}\eta$ smaller for high turbulence cases.
- 2) The temperature and supersaturation spectra (Figs. 1(b)-(c)) do not decay smoothly if phase change takes place (Case H and L) but do so otherwise (Case H0). It appears that there is a surge in the high-wavenumber region before the final decay, particularly for Case L. This surge is driven by the upscale energy transfer caused by phase change (evaporation). The positive slope in the E_{C_d} indicates that phase change injects variance preferentially at small scales ((Fig. 1(d)). The upscale energy transfer is more conspicuous if the turbulence intensity is low (L) since the direct energy cascade is relatively weaker.
- 3) Spectral break is characterized by a sudden change of the slope in the wavenumber space and correspond to the wavenumber k_b . The analogous physical-space break can be calculated as $l_b = (2\pi/k_b)$. Two breaks exist in the E_{q_l} ((Fig. 1(e)) for both cases. For Case H, we calculate that $l_{b,left} = (2\pi/(10^{-1.4}/0.00088)) \sim 139$ mm and $l_{b,right} = (2\pi/(10^{-0.25}/0.00088)) \sim 17$ mm. The corresponding values for Case L are $l_{b,left} = (2\pi/(10^{-1.45}/0.00153)) \sim 271$ mm and $l_{b,right} = (2\pi/(10^{-0.2}/0.00153)) \sim 15.23$ mm.
- 4) If $k > k_{b,right}$, the E_{q_l} is more strongly affected by microphysics than large-scale turbulence. The opposite is true if $k < k_{b,right}$. Between $k_{b,left} < k < k_{b,right}$, the slope is negative for Case L but positive for Case H. So, the influence of microphysics is stronger in the intermediate wavenumber region of E_{q_l} if the turbulence level is low. But flow turbulence is still dominant in the intermediate region for a high turbulence level.
- 5) The peak of E_r is found in the high wavenumber region ((Fig. 1(f)) and reflects the dominance of sub-Kolmogorov fluctuations. The E_r decreases as the k decreases, creating a positive slope, which is typical for an inverse energy cascade.

- 63 6) It appears that the area under the curve is larger for Case L compared to Case H in Figs. 1(b)-
 64 (f). More energy means the magnitude of fluctuations is larger. Higher flow turbulence leads
 65 to stronger mixing and homogenization of the thermodynamic and microphysical gradients in
 66 our model.

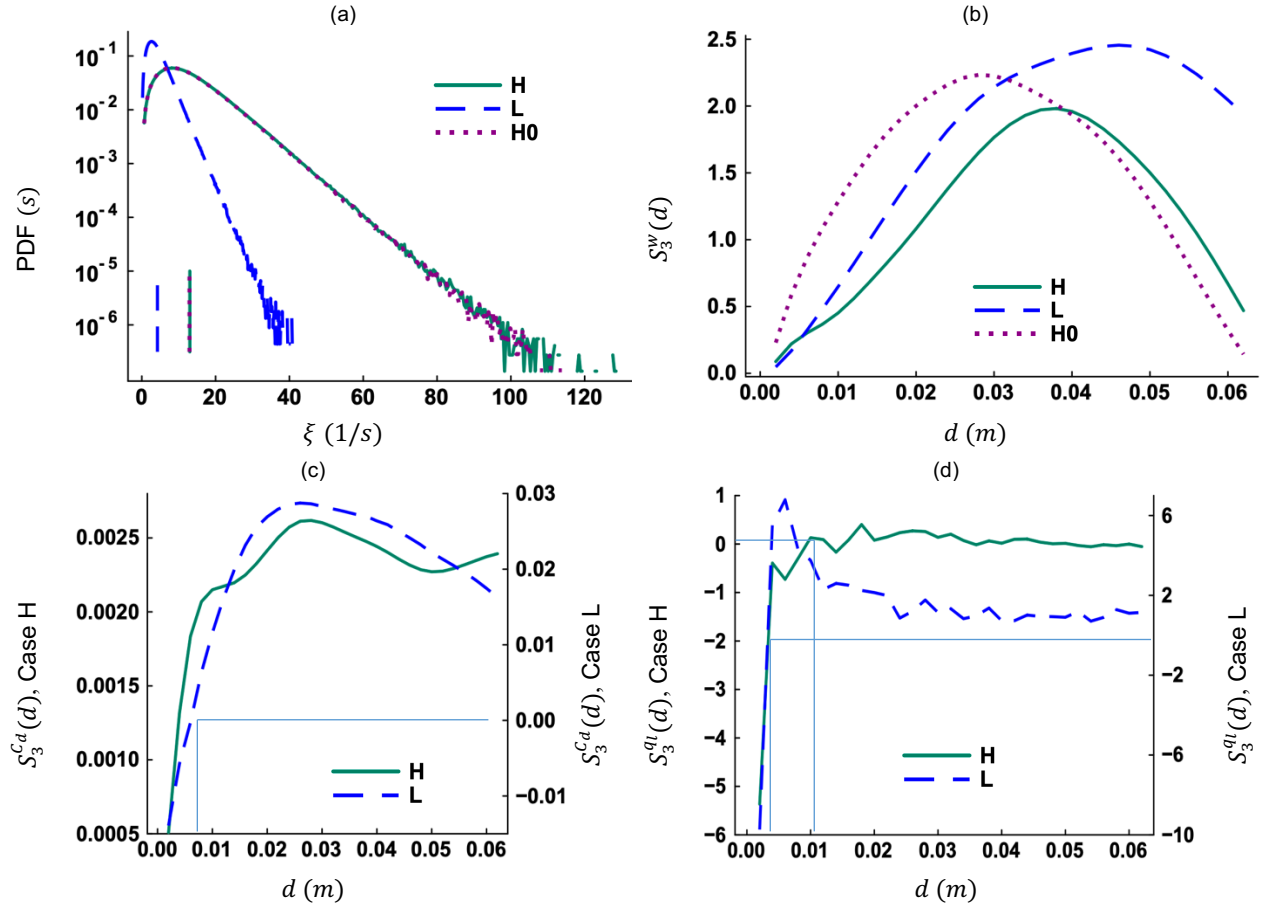
67
 68 The vorticity is calculated as

$$\xi = \left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z} \right) \mathbf{i} + \left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x} \right) \mathbf{j} + \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) \mathbf{k}. \quad (1)$$

69 The compensated third-order structure function for z-velocity is defined as

$$S_3^w(d) = - \frac{\langle (w(\mathbf{x} + \mathbf{d}) - w(\mathbf{x}))^3 \rangle}{\epsilon_{TKE} d}, \quad d = |\mathbf{d}| \quad (2)$$

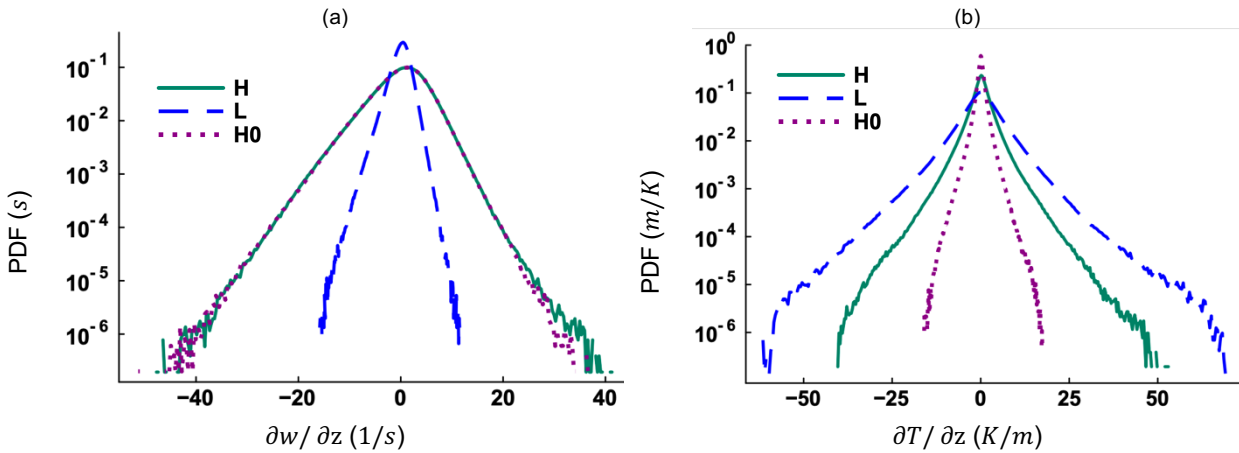
70 where d is the separation distance.



71 **Figure 2.** (a) PDF of the vorticity. Third-order structure functions of (b) z-velocity (S_3^w), (c)
 72 mass exchange term ($S_3^{C_d}$), and (d) liquid water mixing ratio ($S_3^{q_l}$). The straight lines point out
 73 where S_3 is zero in the respective curves.

Summary of findings in Figure 2

- 7) The vorticity PDFs shown in Fig. 2(a) suggest that cloud droplets do not influence the local rotation of the fluid. Expectedly, the mean and maximum vorticities are much higher when the turbulence is stronger. The vorticity PDFs are skewed to the right for all cases, consistent with the vortex stretching phenomenon in homogeneous and isotropic turbulence. Mathematically, the magnitude of vorticity is bounded on the left at zero and any spread of the values is always positively skewed.
- 8) The energy spectra shown in Figs. 1(a)-(f) do not carry directional information. Hence, we plot the compensated third-order structure functions (S_3) in Figs. 2(b)-(d). Note that $S_3 > 0$ implies direct energy cascade while $S_3 < 0$ suggests inverse energy cascade. If $S_3 = 0$, there is no net energy transfer.
- 9) The $S_3^w(d)$ increases with d , reaches a peak, and decays (Fig. 2(b)). The peak value is larger than the Kolmogorov's exact value of 0.8 or $4/5$ for all three cases. However, $S_3^w(d)$ is positive and thus the direct energy cascade is dominant for the z-velocity.
- 10) The $S_3^{cd}(d)$ is much smaller than the $S_3^w(d)$ for Cases H and L (Figs. 2(c)). Interestingly, $S_3^{cd}(d)$ is negative for Case L if d is extremely small. This suggests the existence of inverse energy cascade. The $S_3^{cd}(d)$ increases to zero at $d = 0.007 m$ and is positive as d increases. Thus, direct energy cascade is dominant for C_d while inverse energy cascade is present. This observation is consistent with Fig. 1(d).
- 11) The $S_3^{ql}(d)$ is strongly negative if the separation distance is very small (Figs. 2(d)). Hence, inverse energy cascade is energetically more significant for liquid water mixing ratio at small scales. As d increases, the $S_3^{ql}(d)$ reaches a peak value of 6 and becomes statistically stationary at 1.5 for Case L but does not show a distinct peak and becomes statistically stationary at around 0 for Case H. The magnitude of negative maximum is larger than that of positive maximum for Case L (-10 vs 6) and Case H (-6 vs 0.5). Again, the magnitude of negative maximum is larger for Case L than Case H. This is consistent with Fig. 1(e) as the maximum E_{ql} is found at small scales for each case and is larger for Case L compared to Case H.



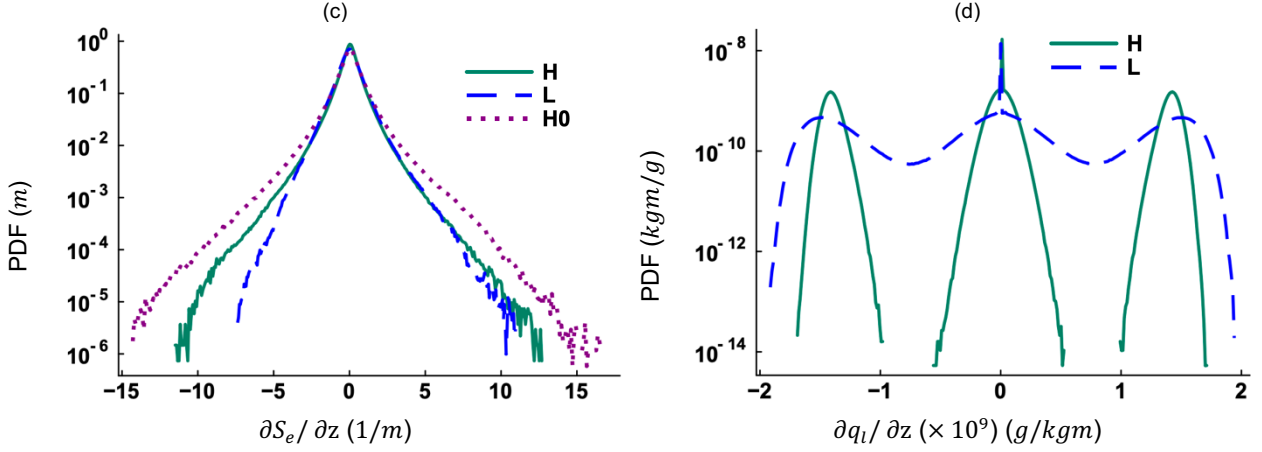


Figure 3. PDFs of the derivatives of (a) z-velocity, (b) temperature, (c) supersaturation, and (d) liquid water mixing ratio. The derivatives are calculated in the z-direction using the central difference formula.

Summary of findings in Figure 3

- 12) Higher turbulence produces larger velocity gradients in Fig. 3(a). The near-perfect overlap for Cases H and H0 again proves that cloud droplets do not modify the velocity field statistics. The distributions are approximately lognormal, meaning the logarithm of the velocity gradient is approximately Gaussian.
- 13) The PDFs of temperature derivatives exhibit tails of the stretched-exponential form for Cases L and H but less so for Case H0 (Fig. 3(b)). Hence, the temperature field is more homogenized if droplets are absent (H0). Phase change generates local gradients which lead to intermittency if turbulence does not smooth them out. Low turbulence means less homogenization and high intermittency.
- 14) The PDFs of supersaturation derivatives also show intermittency caused by phase change (Fig. 3(c)). The spread of the supersaturation PDF follows the order: $H0 > H > L$, which is opposite to that for temperature PDF. This is because phase change acts as a source for temperature gradients but as a sink for supersaturation gradients.
- 15) The PDFs of liquid water mixing ratio derivatives are found to be trimodal for both cases (Fig. 3(d)). The initial domain contained 50/50 cloudy/clear air by volume and subsequent turbulent mixing led to pockets of cloudy and clear air regions without destroying or creating them. The central peak refers to the interior of either the cloudy region (q_l is approximately spatially uniform) or the clear air region (q_l is close to zero). The derivative is zero in both cases. The left peak refers to the cloud side of the cloud/clear-air boundary where q_l transitions from a positive value to zero in clear air. That is, liquid water content decreases in the direction of the gradient. Similarly, the right peak refers to the clear-air side of the cloud/clear-air boundary where q_l transitions from zero to a positive value.

The timescale for large eddy turnover is:

$$\tau_L(\mathbf{x}, t) = \frac{L}{|\mathbf{u}(\mathbf{x}, t)|}. \quad (3)$$

The timescale associated with Taylor microscale is:

$$\epsilon_{TKE}(\mathbf{x}, t) = \nu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) \quad (4)$$

$$\tau_\lambda(\mathbf{x}, t) = \sqrt{\frac{15\nu}{\epsilon_{TKE}(\mathbf{x}, t)}}. \quad (5)$$

136 The Kolmogorov timescale:

$$\tau_\eta(\mathbf{x}, t) = \sqrt{\frac{\nu}{\epsilon_{TKE}(\mathbf{x}, t)}}. \quad (6)$$

137 The timescale for microscale phase change is:

$$\tau_m(\mathbf{X}, t) = \frac{(r(t))^2}{|GS_e(\mathbf{X}, t)|}. \quad (7)$$

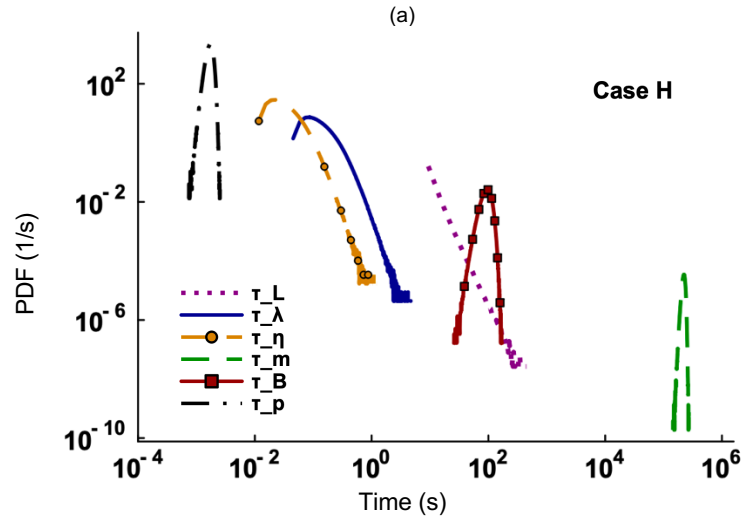
138 The timescale for Brownian diffusion is:

$$\tau_B(\mathbf{X}, t) = \frac{6\pi\mu(r(t))^3}{k_B T(\mathbf{X}, t)}, \quad k_B = 1.38e-23 \quad (8)$$

139 The timescale for droplet response is:

$$\tau_p(t) = \frac{2\rho_l(r(t))^2}{9\rho_a\nu}. \quad (9)$$

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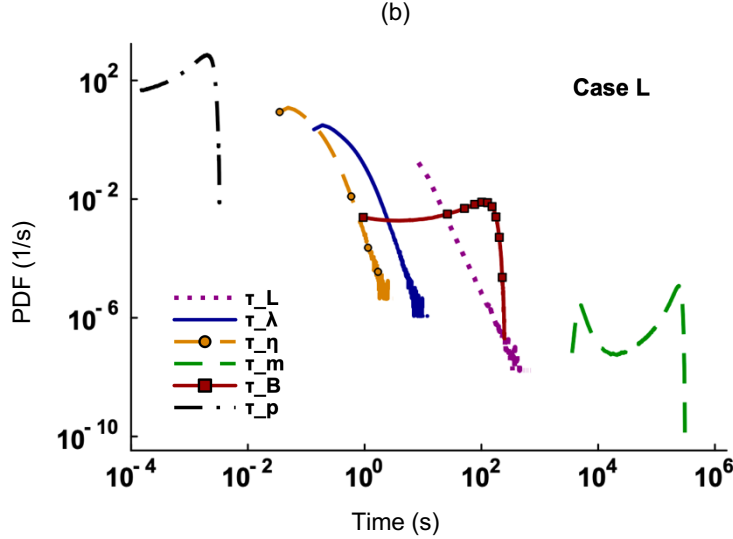


Figure 4. PDFs of the six timescales (τ_L , τ_λ , τ_η , τ_m , τ_B , and τ_p) for (a) Case H and (b) Case L.

Summary of findings in Figure 4

- 16) The six timescales span across 10 orders of magnitude (Figs. 4(a)-(b)). This highlights extreme separation of timescales in turbulent clouds.
- 17) The PDFs of turbulent flow timescales (τ_L , τ_λ , τ_η) are skewed to the right because of turbulent intermittency. It is evident that $\tau_L > \tau_\lambda > \tau_\eta$, which is expected as $L > \lambda > \eta$.
- 18) The droplet response time (τ_p) is much smaller than all three flow timescales for both cases. Hence, the Stokes number (St) is significantly less than unity for every turbulent timescale and droplets are merely tracers in the flow. In particle-laden turbulent flow, turbulence can induce clustering of particles if the latter have significant inertia. Since $St \ll 1$, the probability of clustering is low in our model.
- 19) We see that τ_B is larger than τ_λ and τ_η but overlap with τ_L . Thus, Brownian motion randomizes droplet positions on the same timescale needed by the large eddies to reorganize the flow. This creates competition between two opposing effects on the spatial distribution of cloud droplets. Turbulence continuously moves cloud droplets along correlated fluid pathways, building up spatial correlations between droplet positions. Brownian diffusion is inherently random and tends to destroy those correlations.
- 20) The timescale for microscale phase change (τ_m) is found to be the largest. It means that phase change is quasi-static relative to turbulence and acts as a steady source of q_l variance.

5 Implications of current findings for cloud microphysics

- A) Our particle-based DNS model finds that upscale energy transfer is present in turbulent clouds (Figs. 1(b)-(f) and Figs. 2(b)-(d)). Any parametrization of cloud water mixing ratio (q_l) without accounting for the variance production of q_l at small scales will systematically underestimate cloud microphysics.
- B) During entrainment-mixing, the inverse energy cascade is stronger in weakly turbulent clouds (stratocumulus) than it is in strongly turbulent clouds (cumulonimbus). Thus, spectral break is sensitive to the intensity of flow turbulence (Figs. 1(b)-(f)).

- C) The grid resolution of a cloud microphysics model should be such that it can capture the sign change in the third-order structure function of q_l (Fig. 2(d)).
- D) The trimodal distribution of the derivative of q_l (Fig. 3(d)) suggests that pockets of cloudy and clear-air regions exist within a cloud volume, even though turbulent mixing is going on. Such spatial inhomogeneity is a defining feature of turbulent clouds and cannot be neglected.
- E) The timescales in turbulent clouds differ by up to 10 orders of magnitude (Figs. 4(a)-(b)). The tenfold separation proves that it is very difficult for a single measurement platform to observe cloud physics at all relevant timescales.

References

- Sharfuddin, A.A.M., Ladeinde, F., Liu, Y., Yang, F., Zhang, T., Atif, M., Lin, M. & Lopez-Marrero, V. (2025), Direct numerical simulations of activation and deactivation in turbulent atmospheric clouds. *Physics of Fluids*, 38(1). doi: 10.1063/5.0301098.